$\tilde{\mathbf{E}}$ PHASOR TEMPLATES (Air: $\varepsilon_r = \mu_r = 1$)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ (\text{rad/m}) \quad \Rightarrow \quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table *$u \in \{x, y, z\}$ = axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$\tilde{\mathbf{E}}^i = a_0\hat{e}\, e^{\mp jk_1u}$ (V/m) (incident wave, Med 1)

$\tilde{\mathbf{E}}^r = \Gamma\, a_0\hat{e}\, e^{\pm jk_1u}$ (V/m) (reflected wave, Med 1)

$\tilde{\mathbf{E}}^t = \tau\, a_0\hat{e}\, e^{\mp jk_2u}$ (V/m) (transmitted wave, Med 2)

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x}+\hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x\hat{x} + a_y e^{j\delta}\hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

***Med. 1** ($z < 0$) incident side; **Med. 2** ($z \geq 0$) transmitted.*

	Normal ($\theta_i = 0$)	**$\perp$ pol**	**$\parallel$ pol**
Γ refl. wav	$\frac{\eta_2-\eta_1}{\eta_2+\eta_1}$	$\frac{\eta_2\cos\theta_i-\eta_1\cos\theta_t}{\eta_2\cos\theta_i+\eta_1\cos\theta_t}$	$\frac{\eta_2\cos\theta_t-\eta_1\cos\theta_i}{\eta_2\cos\theta_t+\eta_1\cos\theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2+\eta_1}$	$\frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i+\eta_1\cos\theta_t}$	$\frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_t+\eta_1\cos\theta_i}$
τ rel.	$\tau = 1+\Gamma$	$\tau_\perp = 1+\Gamma_\perp$	$\tau_\parallel = (1+\Gamma_\parallel)\frac{\cos\theta_i}{\cos\theta_t}$
R refl. pwr	$\lvert\Gamma\rvert^2$	$\lvert\Gamma_\perp\rvert^2$	$\lvert\Gamma_\parallel\rvert^2$
T trans. pwr	$\lvert\tau\rvert^2\frac{\eta_1}{\eta_2}$	$\lvert\tau_\perp\rvert^2\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$	$\lvert\tau_\parallel\rvert^2\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$
$R+T$ total	$T = 1-R$	$T_\perp = 1-R_\perp$	$T_\parallel = 1-R_\parallel$

Notes: $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; nonmag. $\Rightarrow$ $\eta_2/\eta_1 = n_1/n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction η_c *$\sigma/(\omega\varepsilon) \ll 1$*

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_r}$. The j term is for $\lvert\eta_c\rvert, \theta_\eta$. See LOSSY MEDIA — 5 CASES.

POWER REFLECTION & TRANSMISSION

% reflected *always* $\%P_r = 100\,\lvert\Gamma\rvert^2$

% transmitted *η-ratio matters!* $\%P_t = 100\,\lvert\tau\rvert^2\,\eta_1/\eta_2$

Check: $\%P_r + \%P_t = 100$.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin\theta_2 = \sin\theta_1\sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin\theta_1\,\frac{n_1}{n_2}$$

Multilayer chain *stack of slabs $1 \to 2 \to 3 \to 4$*

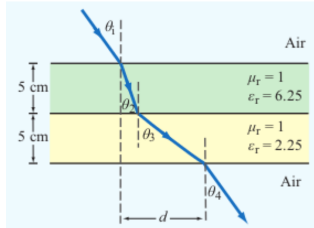
$$\sin\theta_{i+1} = \sin\theta_i\sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes *need Fresnel coefficients (parallel vs perpendicular).*

LATERAL DISPLACEMENT



Beam offset through N slabs *thickness t_i, refracted angle inside slab i* **Slab-indexed** *θ_i = angle inside slab i; t_i = slab thickness (m)*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent *$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon_0 = 8.854\times10^{-12}$ F/m*

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

Type	$\sigma/\omega\varepsilon$	η_c, α, β
Lossless ($\sigma = 0$)	0	$\eta = \eta_0/\sqrt{\varepsilon_r}$ exact $\alpha = 0$, $\beta = \omega\sqrt{\varepsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)$ $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10^{-2}–10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f\mu/\sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep μ_r.

Low-loss quick params *$\sigma/(\omega\varepsilon) \ll 1$, $\mu_r = 1$*

$$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}\ (\text{Np/m}),\quad \beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}\ (\text{rad/m}),\quad \eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega)$$

Exact (quasi-cond) *$X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$*

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1},\quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}$$

$$\theta_\eta = \tfrac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode *$E_z = 0$, $H_z \neq 0$* **TM mode** *$H_z = 0$, $E_z \neq 0$*

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x **form (TE)** *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE$_{mn}$ vs TM$_{mn}$ *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. "a TE wave" $\to$ TE).
2. If $m = 0$ or $n = 0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).
3. Source field $H_z(x,y)$ given $\Rightarrow$ TE. Source field $E_z(x,y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. So TE$_{10}$ exists but TM$_{10}$ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode *generator: $\tilde{H}_z$*

$$\tilde{H}_z = H_0\cos\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{E}_z = 0,\quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TE}},\quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TE}}$$

$$Z_{\text{TE}} = \eta/\sqrt{1-(f_c/f)^2}$$

TM mode *generator: $\tilde{E}_z$*

$$\tilde{E}_z = E_0\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)E_0\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)E_0\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{H}_z = 0,\quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TM}},\quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TM}}$$

$$Z_{\text{TM}} = \eta\sqrt{1-(f_c/f)^2}$$

TEM plane wave *for reference*

$$\tilde{E}_x = E_0\,e^{-j\beta z},\quad \tilde{H}_y = \tilde{E}_x/\eta$$

$$\eta = \sqrt{\mu/\varepsilon}\ (\Omega),\quad f_c\ \text{N/A},\quad k = \omega\sqrt{\mu\varepsilon}$$

Properties common to TE/TM

$$f_c = \frac{u_{p0}}{2}\sqrt{(m/a)^2 + (n/b)^2}\ (\text{Hz})$$

$$\beta = k\sqrt{1-(f_c/f)^2}\ (\text{rad/m})$$

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1-(f_c/f)^2}}\ (\text{m/s})$$

CUTOFF FREQUENCY

Common cutoffs ($a > b$) *f_c formula in Table 8-3*

Mode	f_{mn}
u_{p0}	$c/\sqrt{\varepsilon_r}$ (hollow: $u_{p0} = c$)
TE$_{10}$	$f_{10} = u_{p0}/(2a)$ (dominant)
TE$_{01}$	$f_{01} = u_{p0}/(2b)$
TE$_{20}$	$f_{20} = u_{p0}/a = 2f_{10}$
TE$_{11}$,TM$_{11}$	$f_{11} = \frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$. In Z_{TE}/Z_{TM} and u_g, $f_c = f_{mn}$ of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) *result is unitless*

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]\ (\text{unitless})$$

Inputs: ω (rad/s), β (rad/m), a, b (m), $c = 3\times10^8$ (m/s), m, n (unitless).

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0}\sqrt{1-(f_{mn}/f)^2}\ (\text{m/s}),\qquad u_pu_g = u_{p0}^2$$

$t = L/u_g$ (s) travel time through guide of length L

Higher modes have lower u_g $\Rightarrow$ arrive later.

BOUNDARY

η_1, η_2 — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
$\tau = 1+\Gamma$; $R = \lvert\Gamma\rvert^2$
$k_i = \omega\sqrt{\varepsilon_{r,i}}/c$ (rad/m)
α_2 (Np/m), β_2 (rad/m): lossy
$z = 0$ — boundary plane (m)
$\eta_0 = 120\pi \approx 377\,\Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
$n_i = \sqrt{\varepsilon_{r,i}}$ — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
$\perp$: $\mathbf{E} \perp$ this plane
$\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); $a > b$
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
$Z_{\text{TE/TM}}$ — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
$\eta = \eta_0/\sqrt{\varepsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\varepsilon_r}$ — bulk phase vel.
$u_p = u_{p0}/\sqrt{1-(f_c/f)^2}$
$u_g = u_{p0}\sqrt{1-(f_c/f)^2}$
$u_pu_g = u_{p0}^2$
$t = L/u_g$ — travel time
$c = 3\times10^8$ m/s

POWER & UNITS

$\%P_r, \%P_t$ — pwr refl./trans. (unitless)
$\%P_r = 100\lvert\Gamma\rvert^2$
$\%P_t = 100\lvert\tau\rvert^2\eta_1/\eta_2$
$\%P_r + \%P_t = 100$
σ (S/m), ε (F/m), μ (H/m)
$\varepsilon_0 = 8.85\times10^{-12}$ F/m
$\mu_0 = 4\pi\times10^{-7}$ H/m

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

$$\boxed{\lambda \text{ (wavelength)} = \frac{c}{f} = \frac{3\times10^8}{f} \text{ (m)}}$$

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199\,\Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 kl\eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Power density formulas *$l/\lambda < 1/50$; far field*

Power density	$S(R,\theta) = S_{\max}\sin^2\theta$ (W/m^2)
Max pwr density	$S_{\max} = \dfrac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m^2)

l: rod length (m)	I_0: peak current (A)
R: obs. distance (m)	θ: angle from dipole axis (rad)
$k = 2\pi/\lambda$ (rad/m)	$\eta_0 = 120\pi \approx 377\,\Omega$

Total radiated power $P_{\text{rad}} = \frac{\eta_0\pi}{3}(I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any l*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow$ $S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow$ 0.

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

$^\circ$	rad	$\sin\theta$	$\cos\theta$	$\sin^2\theta$	$\cos^2\theta$
0	0	0	1	0	1
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) *$k = 2\pi/\lambda$, $\gamma = kd\cos\theta$*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\max} = N^2 \text{ at } \theta = \pi/2$$

Normalize $F_a^{\text{norm}} = F_a/N^2$

HPBW (broadside, uniform) *use Nd, not $(N-1)d$*

$$\beta \approx 0.88\frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ\cdot\frac{\lambda}{Nd}$$

Grating lobes appear if $d > \lambda$. Avoid.

BEAM PARAMETERS — β, Ω_p, D

Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Beamwidth β at threshold X

Set $F(\theta_X) = X$, solve for θ_X; $\beta = 2\theta_X$ (symmetric)

Threshold	$X = 10^{\text{dB}/10}$	θ_X (Gauss. shortcut)
HPBW (-3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
-6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
-10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$

Pattern solid angle Ω_p *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi \text{ (sr)}$$

No ϕ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\, d\theta$.

θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F = 1$)	$4\pi \approx 12.57$
Hertzian $\sin^2\theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2/\ln 2$
2-axis HPBW (aperture)	$\beta_{xz}\cdot\beta_{yz}$
Other (use integral)	numerical

TI-36X Pro: `[2nd][d/dx]` for $\int_\square^\square$, RAD mode, multiply by 2π.

Directivity D *always $D = 4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

WORD TRAP: "direction" = $\theta_{\max}$ (rad); "directivity" = D (unitless).

Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p/\lambda^2$. 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D \approx 4\pi/\beta^2$.

Gain (if asked) *ξ = rad. eff., $0 < \xi \leq 1$*

$$G = \xi D \qquad G_{\text{dB}} = 10\log_{10} G$$

Lossless / ideal antenna $\Rightarrow$ $G = D$.

COMMON DIPOLE PARAMETERS

Lossless ($\xi = 1$): $G = D$.

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

ANTENNA AREAS (A_e, A_p)

Effective area *antenna's RX cross section; D from BEAM PARAMS or table above*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section *wire dipole, diameter d; l see table above*

$$A_p = l\,d \text{ (m}^2\text{, any dipole)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).

FRIIS TRANSMISSION FORMULA

For each antenna's G: if given by NAME ("half-wave dipole" etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).

Pwr density at RX, then RX pwr *G_t, G_r linear gains; $\lambda = c/f$*

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad \boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t/L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t\, F_t(\theta_t,\phi_t)F_r(\theta_r,\phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.
3. Get G_t, G_r by antenna type:
- named dipole $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$ lossless)
- aperture / dish $\rightarrow$ APERTURE ANTENNAS ($G = D \approx 4\pi A_p/\lambda^2$)
- arbitrary pattern $F(\theta)$ $\rightarrow$ BEAM PARAMS ($G = \xi\cdot 4\pi/\Omega_p$)
- given in dB $\rightarrow$ dB$\leftrightarrow$LINEAR table

4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G, D, P_r/P_t, SNR, ...):

$$X_{\text{dB}} = 10\log_{10} X_{\text{lin}} \qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow$ $\times 2$, 10 dB $\Leftrightarrow$ $\times 10$. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B T_{\text{sys}} B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m^2); A_e effective area (m^2, $= \lambda^2 D/4\pi$); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ, d aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x\ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x\beta_y)$ $= 4\pi\ell_x\ell_y/\lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2/\lambda^2$

Scaling (any ratio) *$D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$*

$$D_{\text{new}} = D_{\text{old}}\cdot\frac{A_{p,\text{new}}}{A_{p,\text{old}}}\cdot\left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}}\cdot\sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}}\cdot\frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area $\rightarrow$ $D\times 2$, $\beta\times 1/\sqrt{2}$; double freq $\rightarrow$ $D\times 4$, $\beta\times 1/2$.

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3\times10^8$ m/s
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377\,\Omega$
$S(R,\theta)$ — pwr density (W/m^2)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
a — coef. on θ^2 in Gaussian
β — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ rad. eff.; $G = \xi D$
$D_{\mathrm{dB}} = 10\log_{10} D$; $D = 10^{D_{\mathrm{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t/(4\pi R^2)$ (W/m^2)
$P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\mathrm{dB}}/10}$
$3\,\mathrm{dB} = \times 2,\ 10\,\mathrm{dB} = \times 10$
$P_N = k_B T_{\mathrm{sys}} B$ (W)

APERTURE

A_p — physical area (m^2)
$A_e = \lambda^2 D/(4\pi)$ (m^2)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p/\lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D,\ \beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd\cos\theta$
$F_a = \sin^2(N\gamma/2)/\sin^2(\gamma/2)$
$F_a^{\max} = N^2$ at $\theta = \pi/2$
$F_a^{\mathrm{norm}} = F_a/N^2$
$\beta \approx 0.88\lambda/(Nd)$ (broadside)